

Aditya Rao

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EDUCATION

McMaster University

B.Eng in Software Engineering (CO-OP)

Hamilton, ON

Sept 2023 – Present

- 4.00/4.00 CGPA - Dean's Honour List
- Relevant Coursework: Digital Systems, OOP, Discrete Math I & II, Intro to Software Design, DSA, Computer Architecture, Software Development Basics

EXPERIENCE

Software Developer Intern

IBM

May 2025 – Aug 2026

Markham, ON

- Developing WebSphere Application Server v8.5.5, v9.0, WebSphere Liberty, and OpenLiberty and the public and internal tools which support development.
- Re-engineered a suite of 4 internal legacy tools from scratch, modernizing the UX, standardizing Docker deployments, and establishing the first Unit/Integration testing pipelines to eliminate manual verification.
- Created a tool used by over 400+ developers and managers which allows developers to record the distribution of their work they perform and allow managers to record it and use the data to assist the developers by allocating resources in their workflows.
- Optimized the core data processing tool within this suite, replacing heavy framework overhead with custom API wrappers to reduce execution time by 99% (from 10 mins to <4 secs).
- Refactored the critical Product Installation Verification Suite, migrating 20+ legacy FAT tests to a unified framework which enhanced debuggability and system reliability.
- Diagnosed and solved >50 build breaks and supported the development team in diagnosing customer cases and build breaks on all platforms, including but not limited to Linux, Windows, MacOS, IBM i, and zOS
- Created E2E tests for deploying Liberty on OCP (Kubernetes) and verifying behavior along with using third-party software such as Infinispan, Keycloak, and Kafka.
- Created an AI Agent leveraging IBM Bob Shell to analyze build breaks given the build break issue and description, build break logs and server configuration, OpenLiberty and WebSphere Liberty source code reducing time spent on fixing build breaks by at least 2 days worth of analysis.

Full Stack Developer Intern

Medium AI

May 2024 – Aug 2024

Hamilton, ON

- Developed and maintained React and TypeScript code for Medium AI, contributing to 2 first-place and 1 second-place finishes in the McMaster Startup Survivor competition, and helping secure \$10,000 in funding.
- Implemented real-time speech transcription using Python, OpenAI Whisper, FFmpeg, and Socket.IO

PROJECTS

WebCraft | 🌐 | C++, CMake, Vcpkg, Asynchronous Programming, Github Actions

Apr 2025 – Sept 2025

- Developed an asynchronous runtime and programming library using C++ 23 features to allow developers who are used to Node.js, Python, and Java to write fast and simple async C++ code easily.
- Created coroutine abstractions, the async runtime, async streams, file and networking module abstractions over low level OS specific async library code to simplify cross-platform C++ development.
- Created CI/CD pipelines and created over 130+ tests to verify correctness using gtest, CTest, Vcpkg, and Github Actions to ensure code quality and reliability.
- Created a vcpkg port so developers can easily access it.

Resume Builder | 🌐 | Java, Spring Boot, Next.js, Typescript, Python, FastAPI, LangChain, Docker

Aug 2024 – Feb 2025

- Developed an AI-powered resume builder for automated resume creation, scoring, and real-time editing.
- Improved resume management by reducing manual effort in tailoring resumes for job applications.

FashioNova | 🌐 🌐 | Typescript, Next.js, Python, Socket.io, OpenCV, Cloudflare AI, Flask

Oct 2024 – Oct 2024

- Developed a virtual wardrobe and AI fashion advisor in 36 hours at Hack the Valley 9 at UofT Scarborough
- Created an OpenCV pipeline using Python Flask and Socket.io to fit virtual clothes onto users via live video feed
- Leveraged Cloudflare AI to suggest outfits based on the occasion, promoting sustainable fashion consumption.

Smart Study Buddy | 🌐 🌐 | Python, Flask, Wolfram Alpha API, OpenAI API, Next.js, Typescript

Oct 2024 – Oct 2024

- Developed an intelligent chatbot capable of summarizing course materials and answering academic queries.
- Integrated OpenAI API for natural language processing to provide general question-answering capabilities.
- Utilized Wolfram Alpha API to handle scientific and computational queries, offering detailed responses.
- Built a feature to generate practice problems based on course material and enhance learning using Python and Flask.

QuakeGuard | 🌐 | *Next.js, Flask, Typescript, Python, TensorFlow, Keras, SciKit Learn, Pandas* Jan 2024 – Jan 2024

- Developed an earthquake prediction tool using ML models showcased at DeltaHacks X and GDSC McMaster
- Visualized global earthquake data and high-risk areas with Next.js and Python, Flask, SciKit Learn, and TensorFlow

EXTRA CURRICULARS

Sys-Admin and Fullstack Developer Executive

Nov 2024 – Present

IEEE McMaster Student Branch

Hamilton, ON

- Set up an RPi 3B+ Cluster for a server to handle up to 100 clients in an instant with a budget of \$5000
- Managed and provisioned a CI/CD pipeline which generates Docker images for our Kubernetes cluster
- Developing the website for the IEEE student branch and the admin dashboard for executives to manage

TECHNICAL SKILLS

Languages: Java, Python, SQL, .NET Framework, C, C++, JavaScript/TypeScript, Verilog

Frameworks: Spring Boot, Jakarta EE, MicroProfile, Hibernate, Express, Next.js, FastAPI, Flask, UWP

Developer Tools: VS Code, IBM Bob, Eclipse, AWS, GCP, Git, Github, Docker, OpenShift, Kubernetes, Databases, Infinispan

Libraries: Pandas, TensorFlow, OpenCV, SciKit Learn, JavaFX, Redux, JQuery, React, Mongoose

Hobbies: Sketching, Painting, Violin (Western and Carnatic), Badminton, Swimming, Chess

CERTIFICATIONS

Clean Code

September 2025

Udemy

IBM Agile Explorer

September 2025

IBM

The DevOps Essentials - The Handbook

September 2025

Udemy

Introduction to Containers, Kubernetes, and OpenShift

August 2025

IBM

Practical OpenShift for Developers - OpenShift 4

August 2025

Udemy

[NEW] Ultimate AWS Certified Cloud Practitioner CLF-C02 2025

February 2025

Udemy

Docker & Kubernetes: The Practical Guide [2025 Edition]

January 2025

Udemy